

Development and Implementation of a Loyalty System for a Confectionery Based on Smart Contracts

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INTRODUCTION: PROBLEM STATEMENT

Traditional loyalty systems lack transparency, interoperability, and trust.

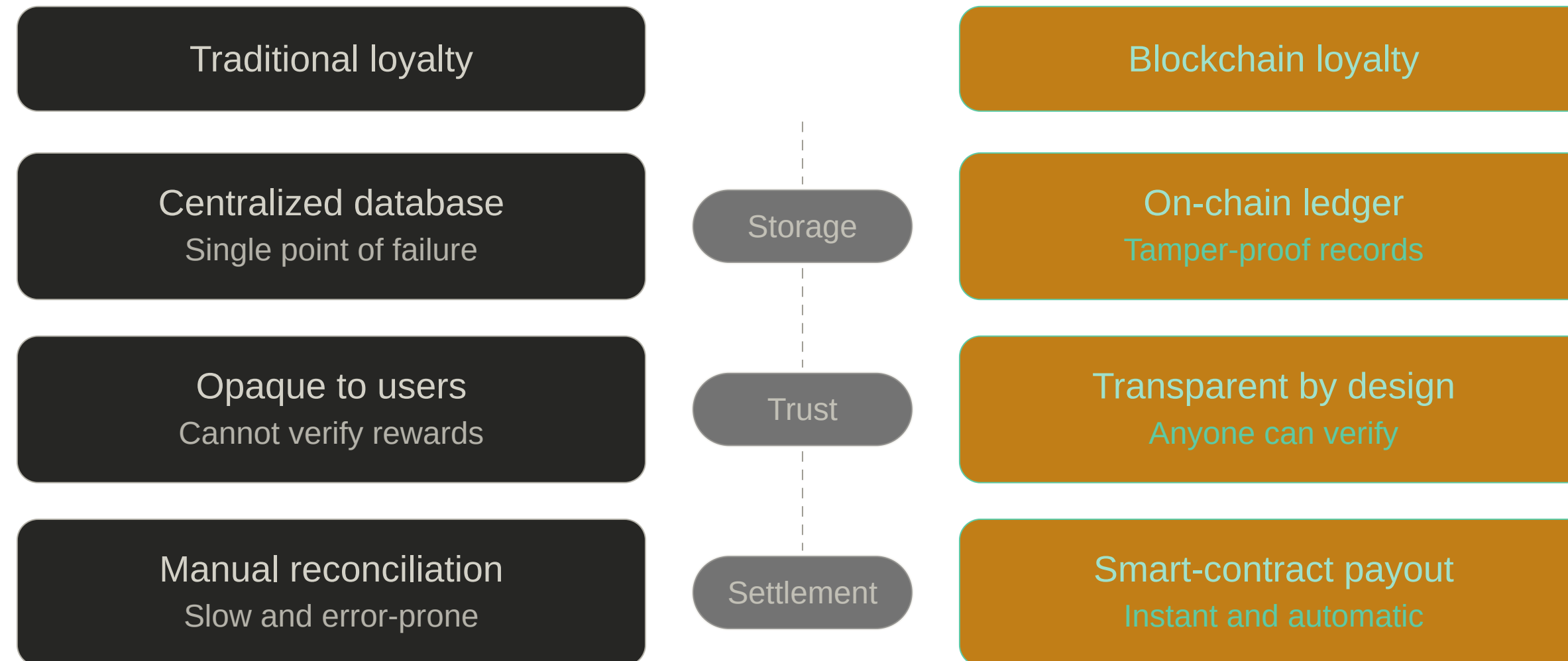


Figure 1: Comparison of traditional and Blockchain-based loyalty systems.

Challenges of traditional loyalty systems:

- Centralized reward storage
- Limited transparency
- Risk of data manipulation
- Difficult partner integration
- High administrative effort

How to make loyalty programs transparent, secure, and user-friendly?

Blockchain technology enables transparent, auditable, and tamper-resistant reward management.

INTRODUCTION

AIM

Develop a blockchain-based loyalty platform for confectionery businesses using smart contracts and Telegram Mini Apps.

OBJECTIVES

- Analyze existing loyalty solutions •
- Design system architecture •
- Develop smart contracts •
- Implement Telegram Mini App •
- Evaluate customer and business benefits •

RELEVANCE

- “55% of loyalty members don't know their points balance, and 41% are unaware of their points value”. (Bond Brand Loyalty, 2017)
- “Acquiring a new customer costs 5–25 times more than retaining one, driving high administrative overhead”. (Plaisir, M. L., & Team, A., 2024)
- “Only 48.6% of earned loyalty points are ever redeemed, customers can't independently track or verify them” (Garai T., 2023)

Businesses need transparent, trustworthy, and cost-effective loyalty solutions that improve customer engagement and retention.

NOVELTY

- TON blockchain integration •
- Telegram Mini App interface •
- Gasless transactions through relay service •
- Hybrid on-chain/off-chain architecture •

LITERATURE REVIEW

Key Findings from Existing Research:

1. **Traditional Loyalty Systems:** Limited reward interoperability
 2. **Ethereum Dominance:** Most solutions use EVM
 3. **Web3 Usability Issues:** Wallets and fees required
 4. **High User Friction:** Complex customer onboarding
 5. **Hyperledger Adoption:** Better governance and privacy
 6. **Consortium Limitations:** Reduced openness and flexibility
 7. **Research Focus Gap:** Ethereum and Hyperledger centric
- (Sönmeztürk et al., 2020; Nguyen et al., 2021; Wang et al., 2019; Tu et al., 2022)

No existing work combines:

- Public asynchronous blockchains (TON),
- hybrid on-chain/off-chain architecture,
- low-friction B2B loyalty workflows.

Solution Type	Main Limitation
Ethereum loyalty systems	gas fees, scalability
Hyperledger consortium systems	governance overhead
NFT/Web3 loyalty apps	weak everyday utility
Traditional loyalty platforms	centralized databases

Research Gap

Lack of implementation-oriented architectures for scalable and user-friendly blockchain loyalty ecosystems.

PROPOSED SOLUTION

We propose a B2B loyalty ecosystem on the TON blockchain to resolve market fragmentation and enhance customer psychological ownership by leveraging transparency of blockchain operations.

- Transition loyalty points from corporate liabilities into liquid digital assets that customers truly own through tokenization (Wang et al., 2019; Santos et al., 2023).
- Establish a generic, "trust-free" platform where multi-brand rewards coexist in a single user interface to eliminate siloed applications (Santos et al., 2023).

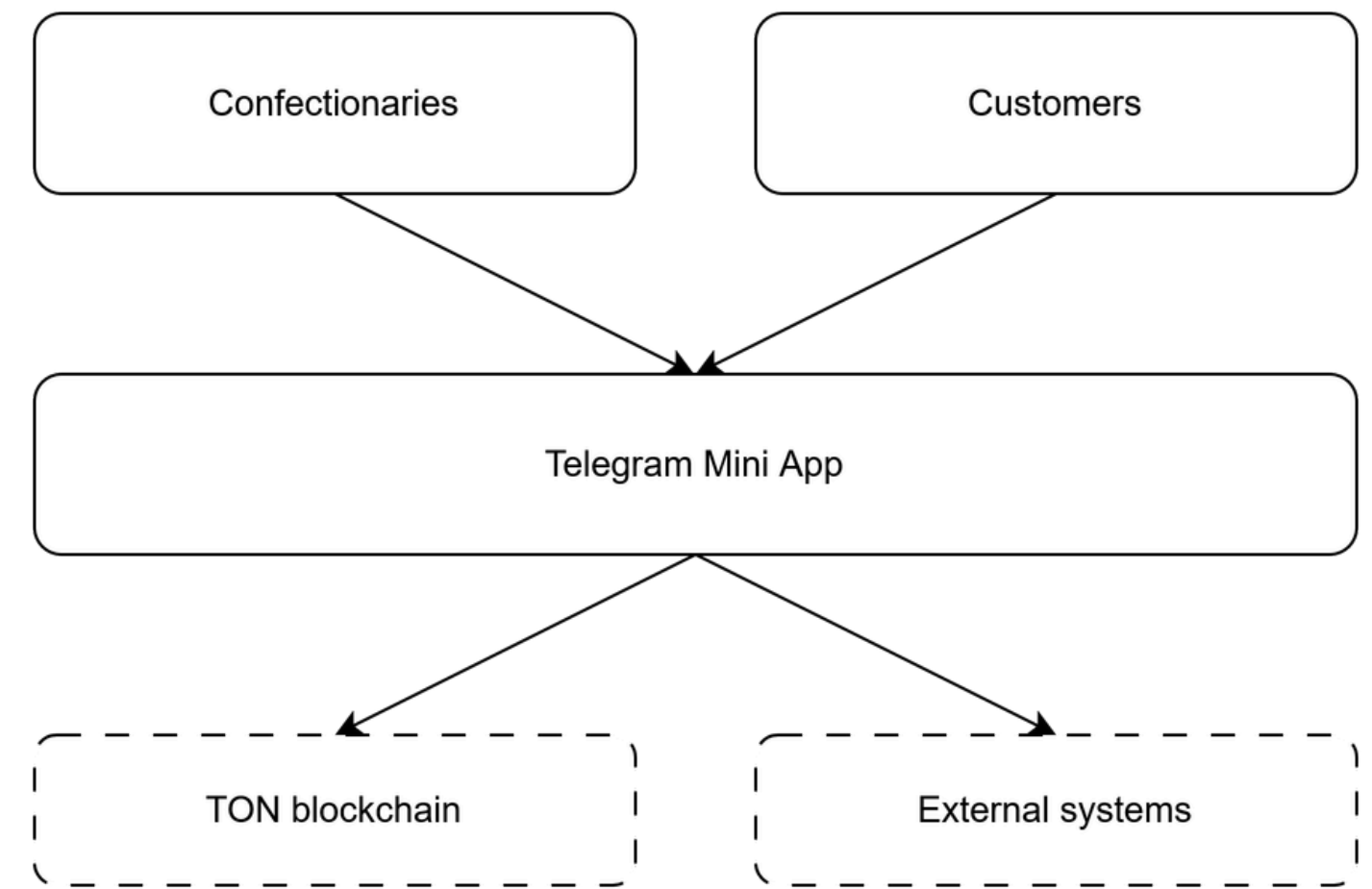
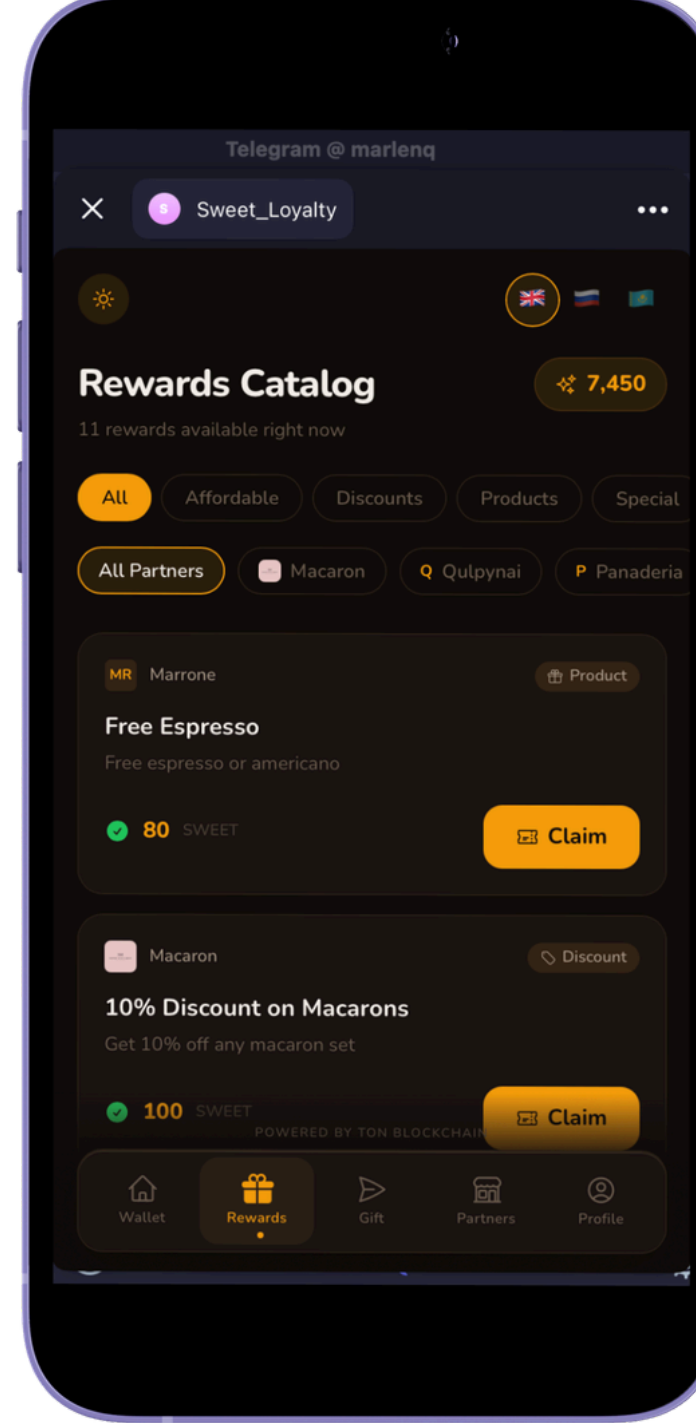
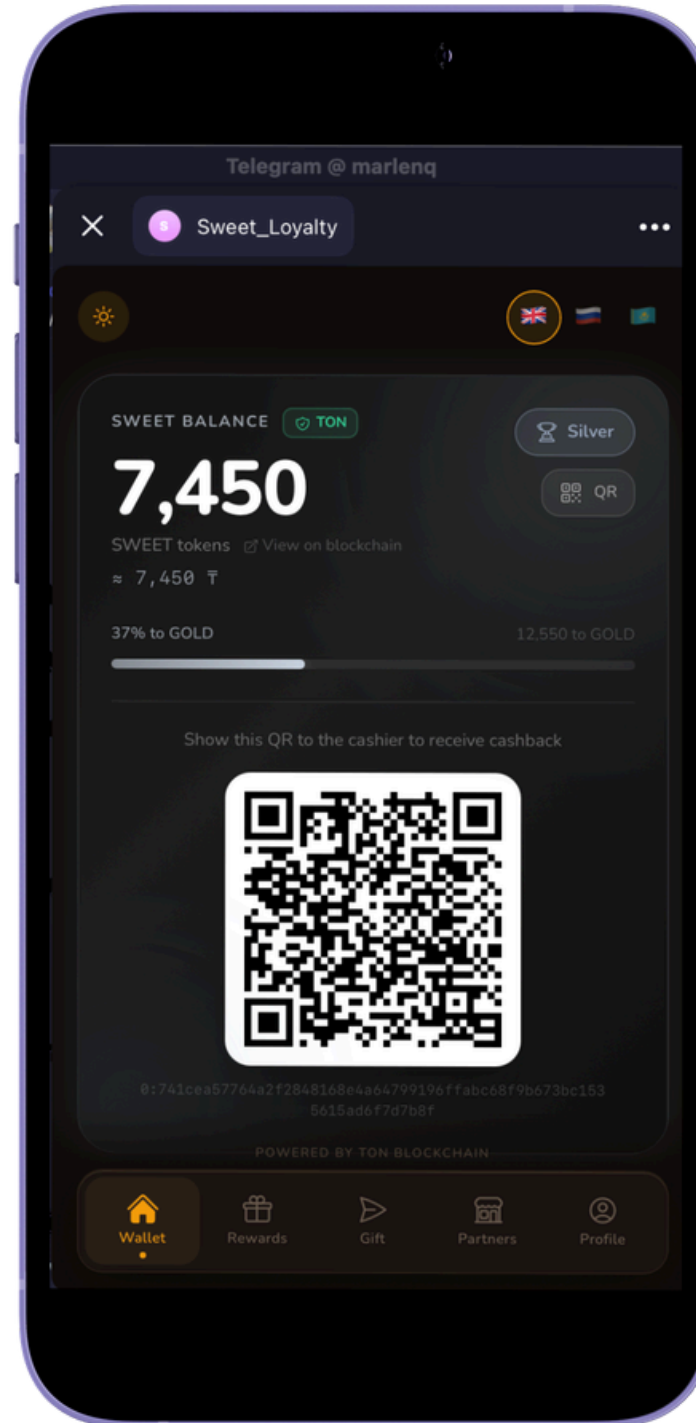
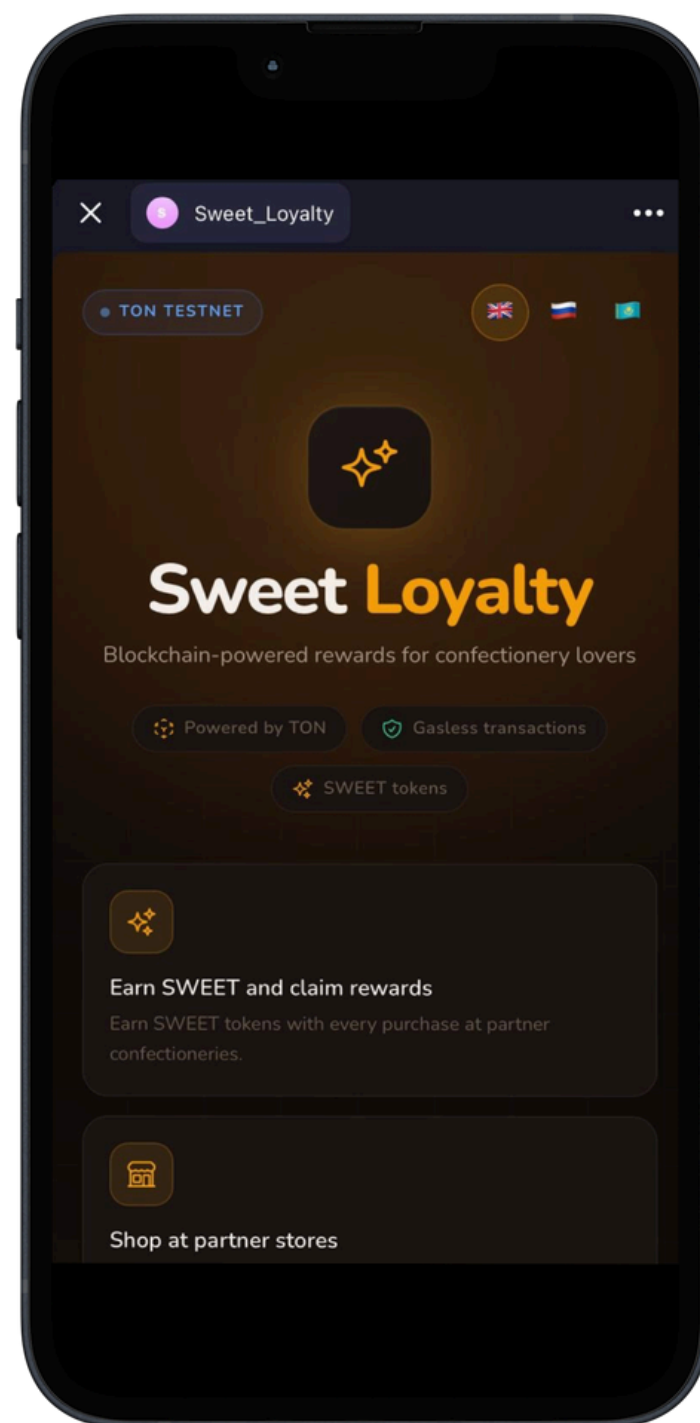


Figure 2: A high level abstraction of a TON-based confectionery loyalty platform

PROPOSED SOLUTION

Sweet Loyalty Platform



Key Features:

- Blockchain reward tracking
- Automated smart contracts
- Telegram-based access
- Cross-partner ecosystem
- Transparent transactions
- MVP-oriented architecture

METHODOLOGY

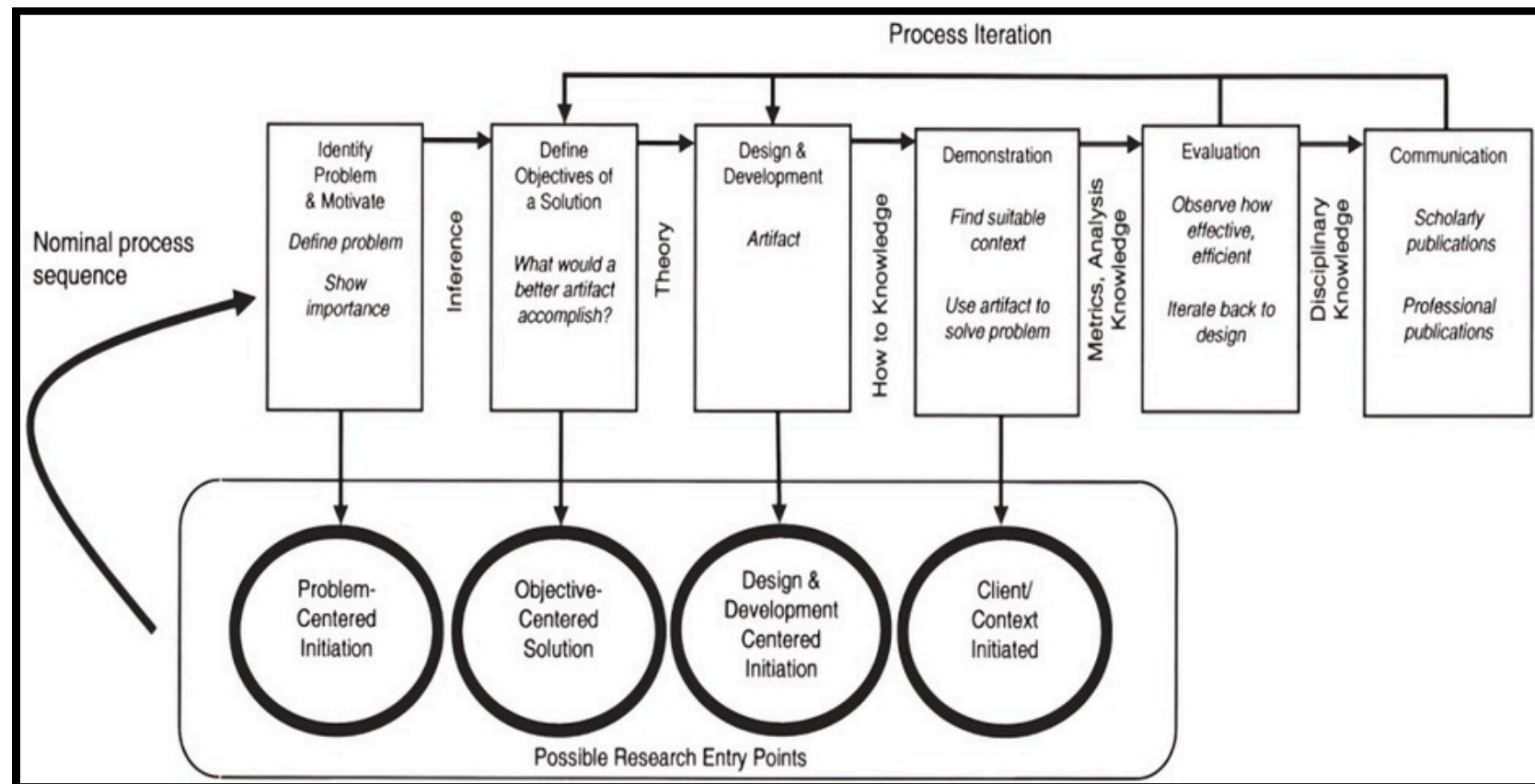


Figure 3: The Design Science Research (DSR) cycle used for the iterative development and evaluation of the TON loyalty artifact (adopted from Behrouzi, 2023)

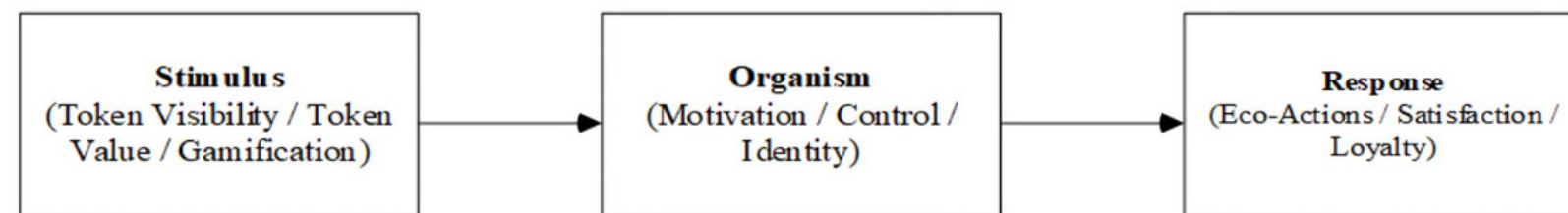


Figure 4: The Stimulus-Organism-Response (SOR) Model of current blockchain-based loyalty programmes (adopted from Behrouzi, 2023).

DSR Framework: A cyclic process moving from problem identification to technical artifact design and expert evaluation. structured architecture development; iterative design and evaluation process.

SOR Model: Modeling how tokenized stimuli (visibility and value) trigger internal states of perceived control to produce loyalty responses. explains behavioral impact of tokenized rewards; models trust and engagement formation.

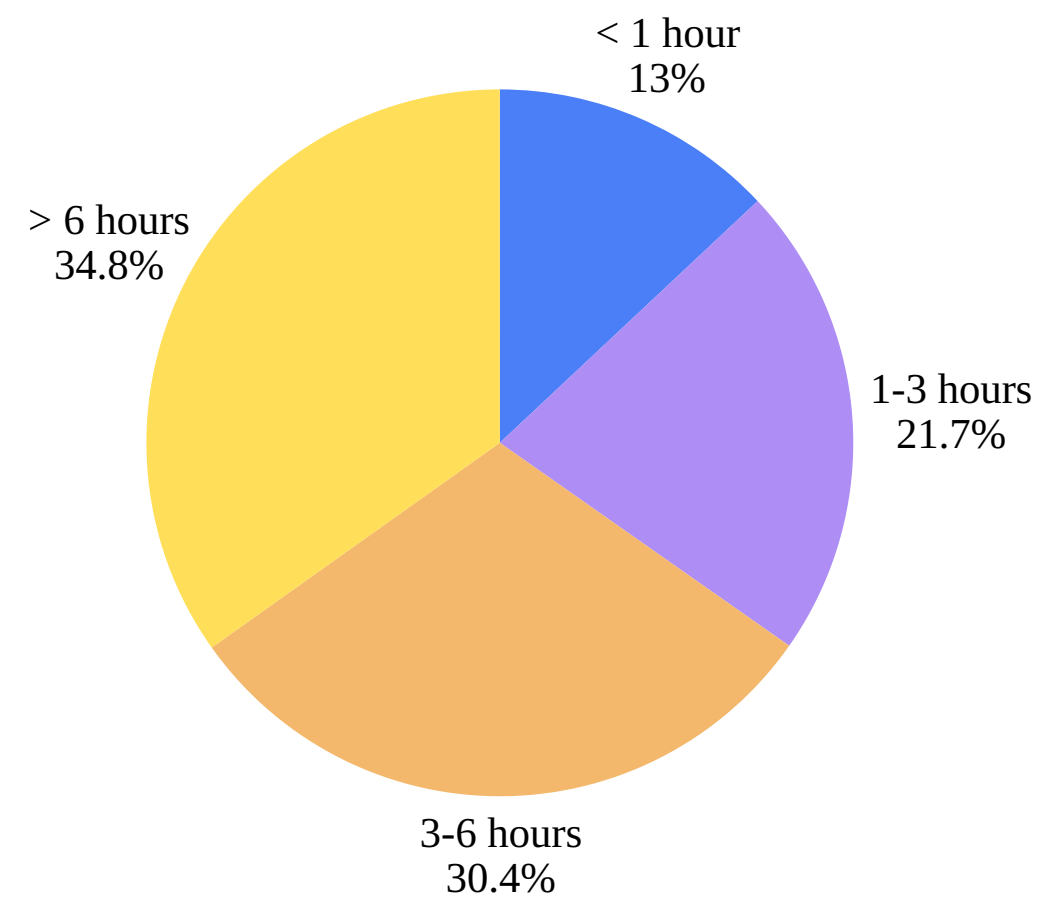
The evaluation focuses on key stakeholder friction points:

- Customer Perception: Surveys on trust and usability.
- Business Perception: Partner feedback on operational friction.
- Financial Feasibility: Transaction volume and payback calculations.
- Architectural Acceptance: Testing readiness via system performance.

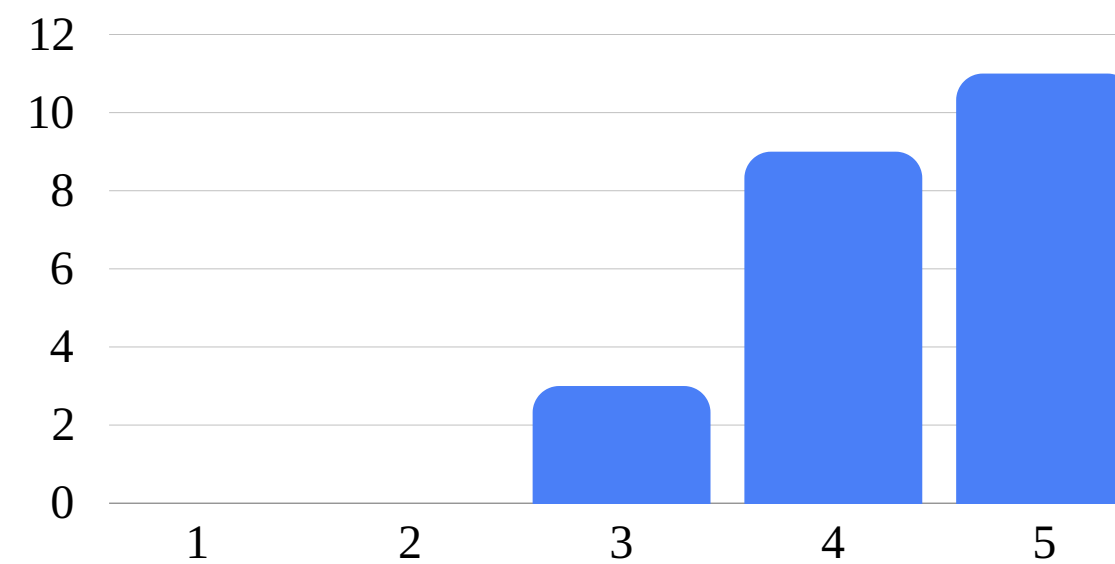
Combines reproducible data transformation, survey-based evaluation, KPI analysis, and architectural assessment. All transformations are explicit and reproducible, ensuring consistent results under identical conditions.

METHODOLOGY

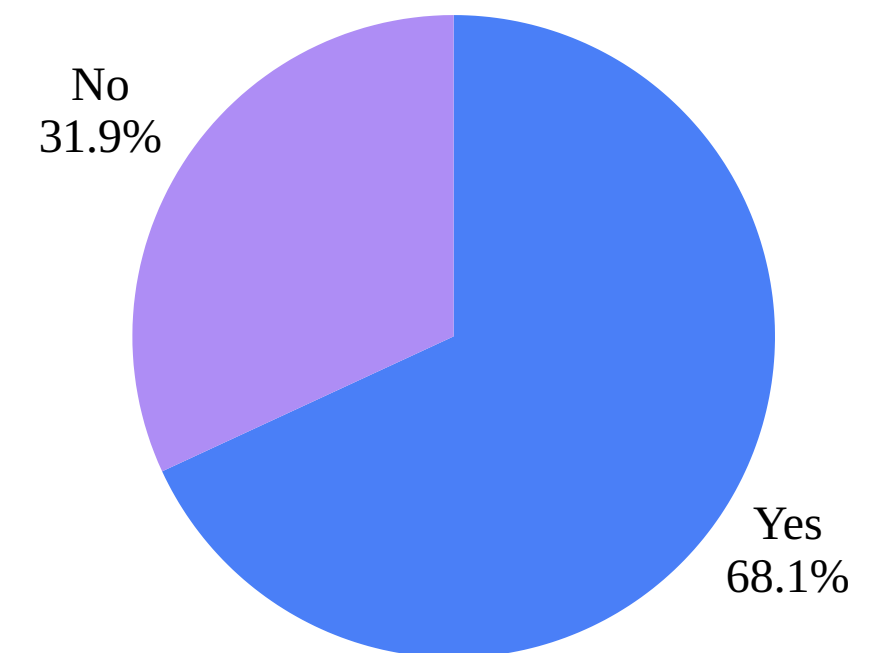
B4. How many hours per week does your team spend on manual loyalty reconciliation or reporting?



B9. Would an auditable blockchain transaction log reduce reconciliation disputes with partners?



C11. Have you ever lost points due to program changes or expiry without notice?



TECHNOLOGY STACK

BACKEND

Node.js | TypeScript | Express.js | PostgreSQL | Prisma | Redis

FRONTEND

React | TypeScript | Vite | TailwindCSS | TonConnect UI

BLOCKCHAIN LAYER

FunC | TON Core | TON Blueprint

DevOps

Vercel Deployment | Docker | Docker Compose | GitHub Actions



ARCHITECTURE

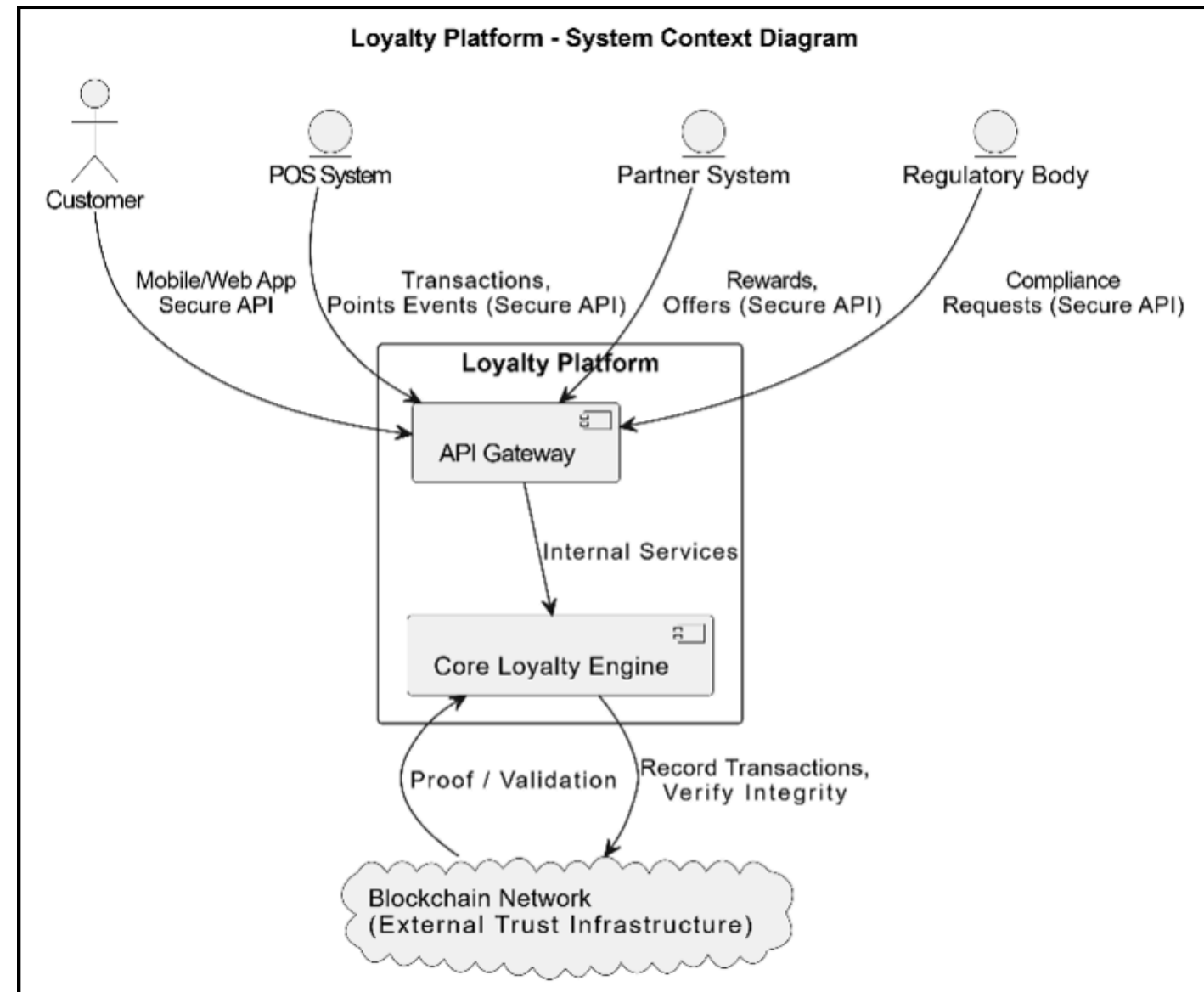


Figure 5: Confectionery system context diagram for the proposed loyalty platform.

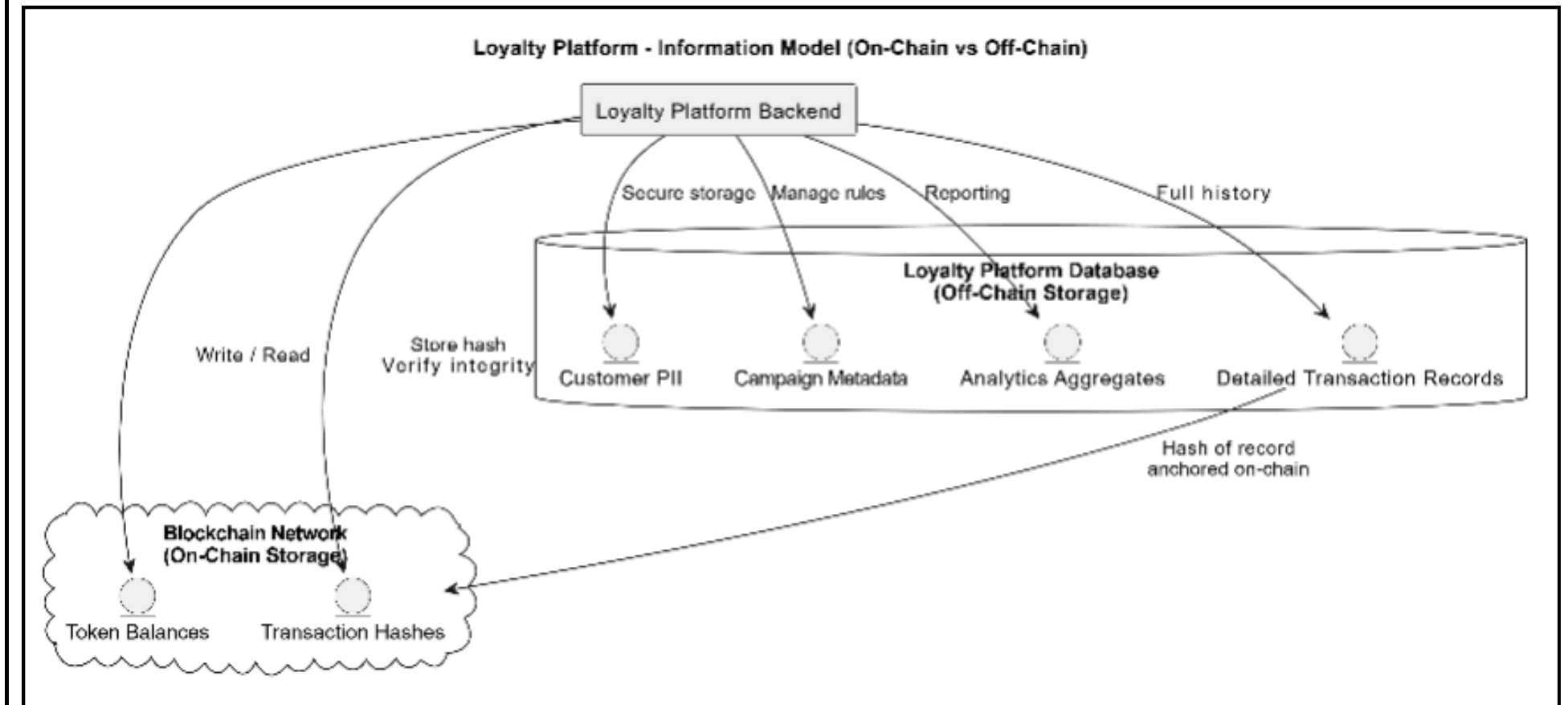


Figure 6: Off- and on-chain informational model of the proposed loyalty platform.

RESULTS

Average net benefit per business	159,871.74 KZT/month
Average payback period per business	7.51 months
Pilot reconciliation hours saved	268.46 hours/month
Pilot total net benefit	3,677,050 KZT/month

Table 1: Aggregate business KPI model summary.

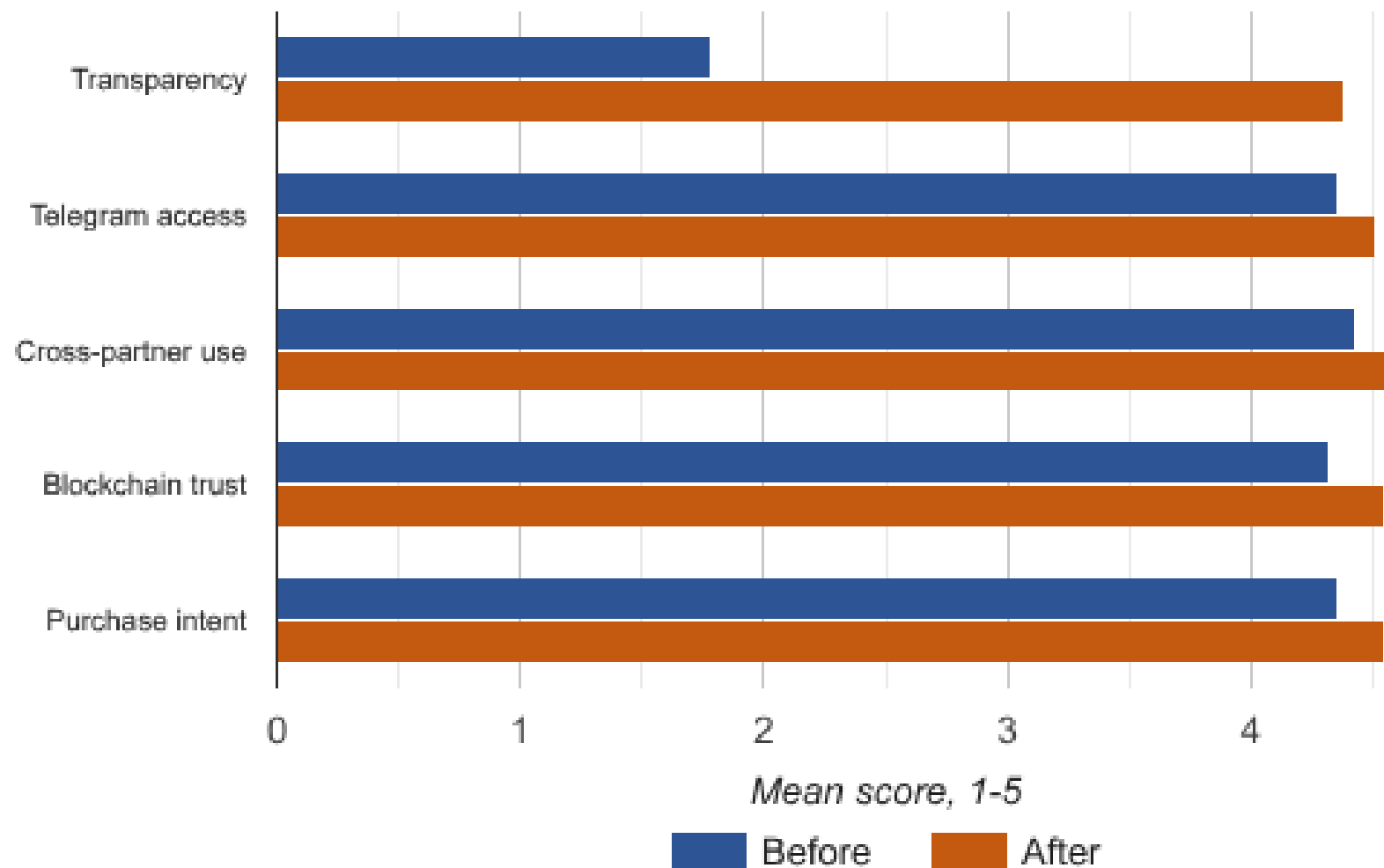
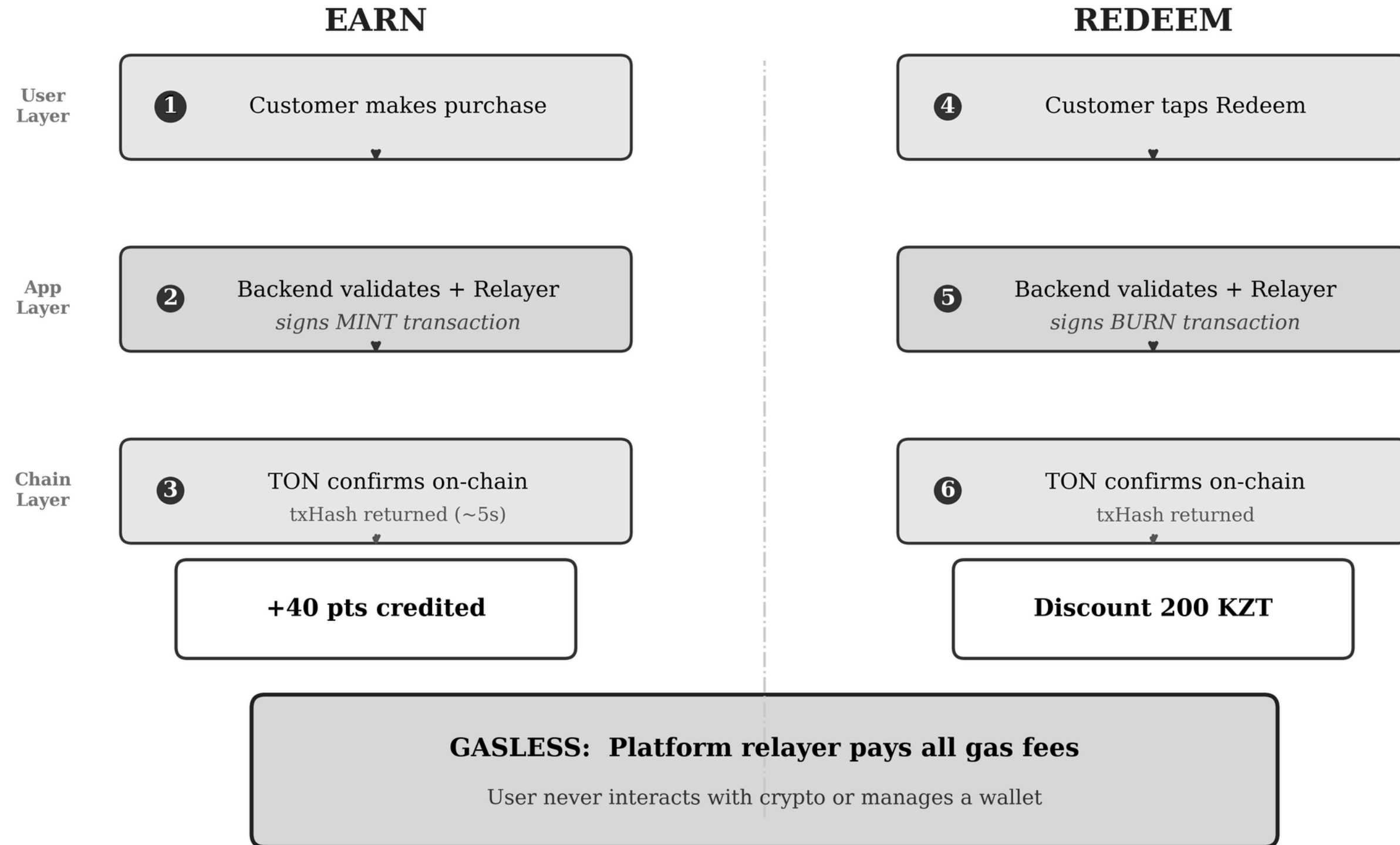


Figure 7: Customer value perception after MVP demonstration

Metric	Before	After	Change
Shared platform	83%	95%	+12pp
Dispute resolution	86%	96%	+10pp
Telegram channel	86%	96%	+10pp
Rule updates	100%	100%	—

Table 2: Business observations before and after pilot

MVP Demonstration Flow



CONCLUSION

This thesis developed a TON-based blockchain loyalty platform for B2B confectionery ecosystems. The proposed architecture combines smart contracts, a Telegram Mini App, and off-chain storage to improve reward transparency, simplify user interaction, and automate partner settlement. The results demonstrate the architectural feasibility of a coalition loyalty system built on TON.

LIMITATIONS AND FUTURE WORK

The study was evaluated at the MVP level using a small regional sample from Astana and Almaty. Production deployment, smart contract auditing, and large-scale performance testing were outside the scope of this research.

Future work includes TON mainnet deployment, independent smart contract audits, expansion to other retail sectors, and DAO-based governance for platform management. Further research should evaluate long-term customer engagement, expand the architecture beyond confectionery businesses to other retail sectors.

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